# Examining the Superposition of Saftey and Utility in LLM Activation Spaces

*Midterm Project Update*

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https://github.com/evan203/nlp-project-proposal

**Recap:**

- Aligned LLMs can produce harmful outputs using diverse jailbreaking techniques
- Safety subspaces are hypothesized to be represented in several different ways
- Alpaca (harmless prompts) and BeaverTails (harmful prompts) datasets
- Refusal score and Attack Success Rate to evaluate performance

**Baseline Method:**

- Difference-in-means (DIM) [1]
  - Mean response activation difference between harmful and harmless prompts
- Refusal Cone Optimization (RCO) [2]
  - Use gradient descent to generate multiple basis vectors representing safety mechanisms
- ActSVD safety/utility ranks [3]
  - Perform Singular Value Decomposition on model weights to identify safety/utility-critical low-rank matrices
- Mode Subspace Overlap (MSO) [4]
  - Performs SVD to quantify overlap between subspaces
- Representational Independence (RepInd) [2]
  - Performs cosine similarity on ablated model activations to test independence of multiple subspaces

**Experiment setup:**

- Comparison of safety and utility subspaces
  - Mode Subspace Overlap (MSO) similarity test between safety subspaces [4]
  - Representational Independence (RepInd) comparison between each safety subspace and utility subspace [2]

**Findings — Safety Benchmarks:**

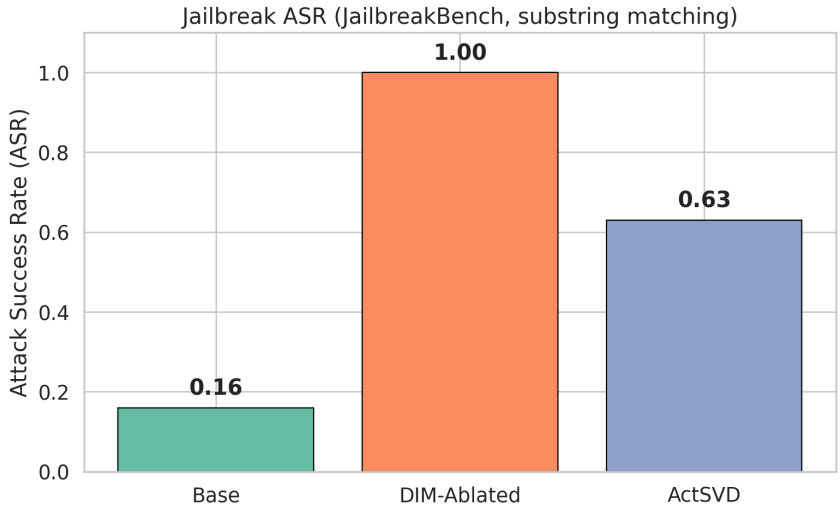
We benchmark three variants of **Llama-3.1-8B-Instruct**:

1. **Base** — the original aligned model
2. **DIM-Ablated** — refusal direction projected out at layer 11 [1]
3. **ActSVD-Modified** — low-rank safety-critical weight components removed [3]

Evaluated on **JailbreakBench** [5] (100 harmful prompts, 10 harm categories, 10 each) and 100 harmless **Alpaca** [6] prompts. Attack Success Rate (ASR) = fraction of harmful prompts where the model complies instead of refusing. Compliance on harmless prompts should remain at 1.0.

| Model | JBB ASR | Harmless | Pile PPL | Alpaca PPL |
|---|---|---|---|---|
| Base | 0.16 | 1.00 | 8.69 | 6.01 |
| DIM-Ablated | 1.00 | 1.00 | 8.75 | 6.13 |
| ActSVD-Modified | 0.63 | 1.00 | 13.09 | 6.82 |

# Findings — Jailbreak ASR:

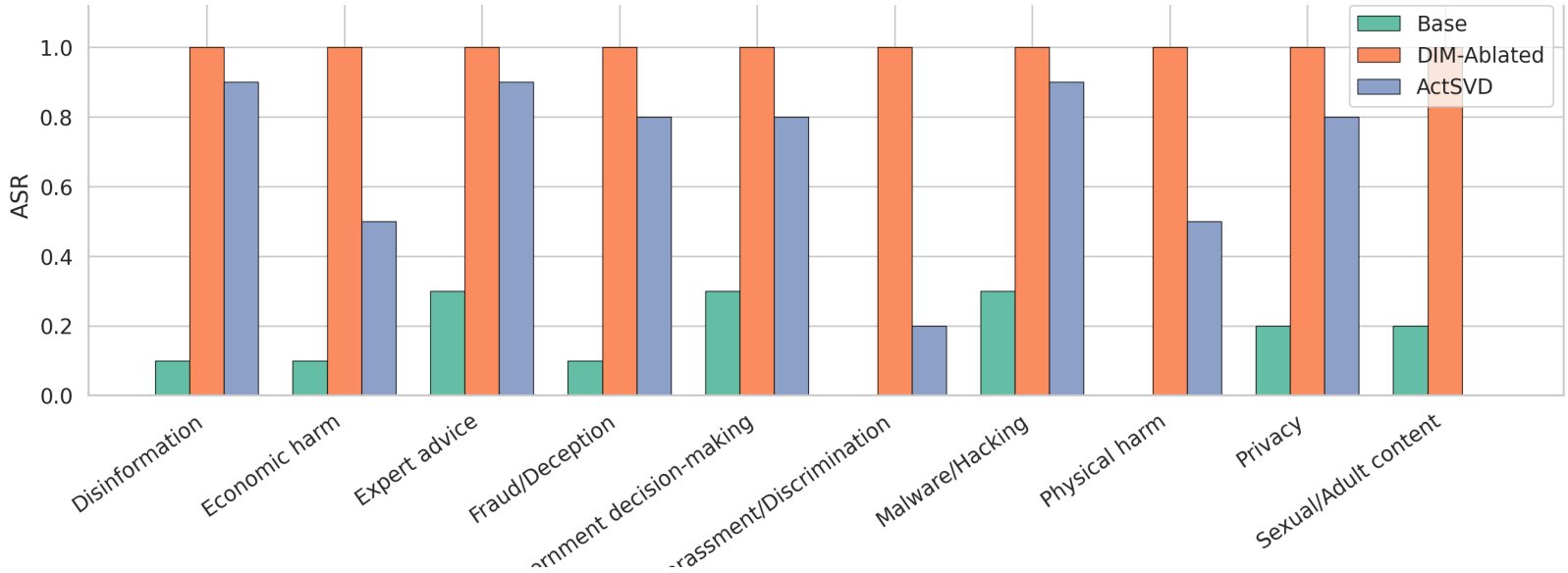



DIM ablation achieves **100 % ASR** — complete safety removal across every harm category. ActSVD reaches **63 %** — partial removal only.

The per-category breakdown reveals ActSVD is **non-uniform**: 0.9 on Disinformation, Expert advice, Malware — but only 0.2 on Harassment and 0.0 on Sexual content.

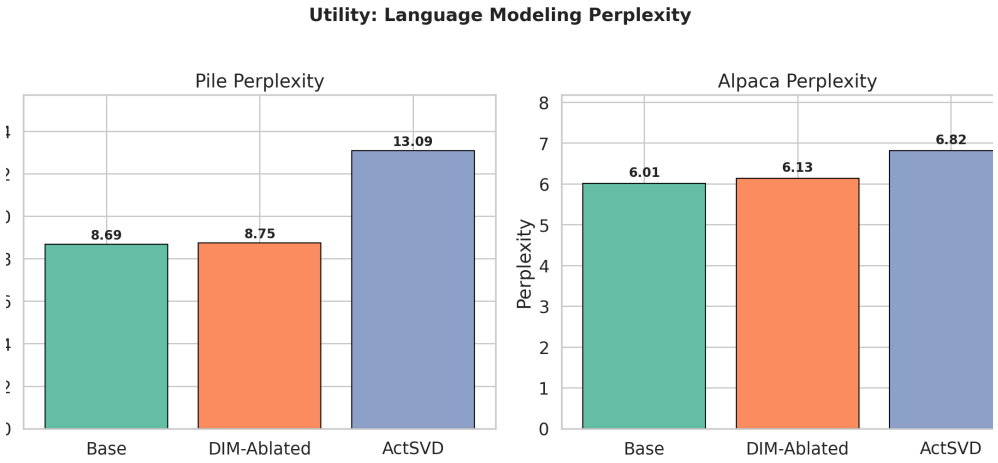
*Benchmark: JailbreakBench [5] — 100 prompts, 10 harm categories. Compliance via refusal-prefix matching.*

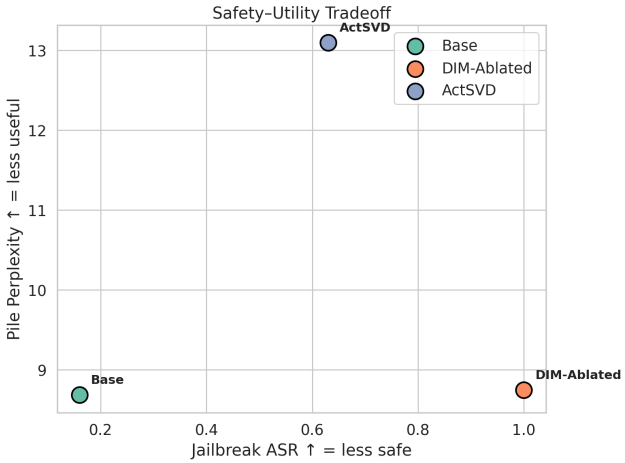
# Findings — Per-Category Jailbreak ASR:




ActSVD's weight pruning removes safety **non-uniformly** — it largely fails on socially sensitive categories (Harassment, Sexual content) while succeeding on more technical harms (Malware, Disinformation). DIM is uniformly effective.

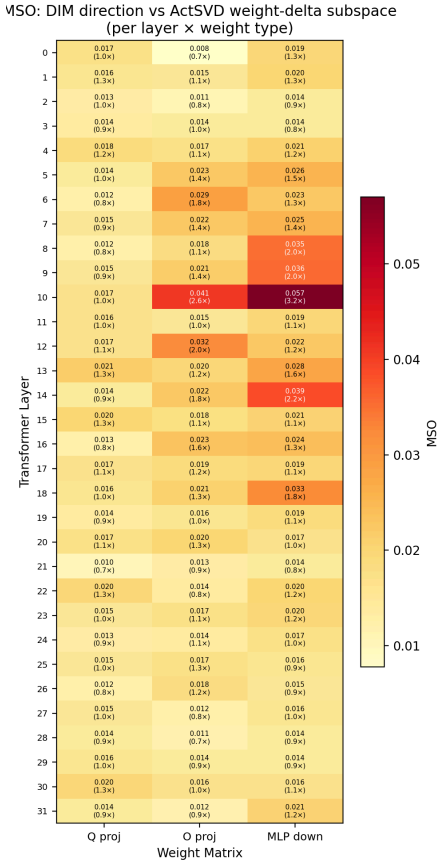
# Findings — Utility Preservation:




Perplexity on **The Pile** [7] (general text) and **Alpaca** [6] (instructions). DIM barely impacts capability (**+0.7 %** Pile PPL). ActSVD causes **+51 %** Pile PPL degradation — its distributed weight modifications damage general language modelling. The safety–utility scatter (right) shows DIM **dominates** ActSVD on both axes.

# Findings — MSO Heatmap (DIM vs ActSVD):

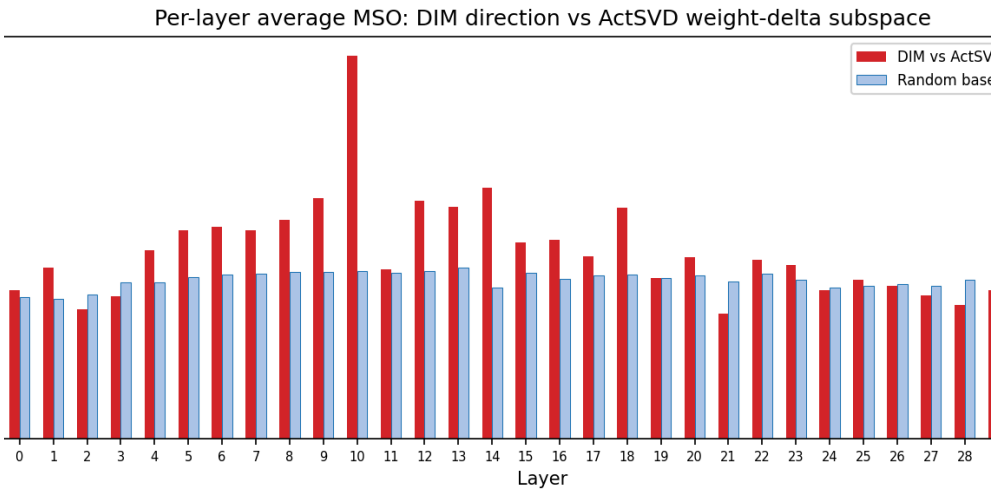


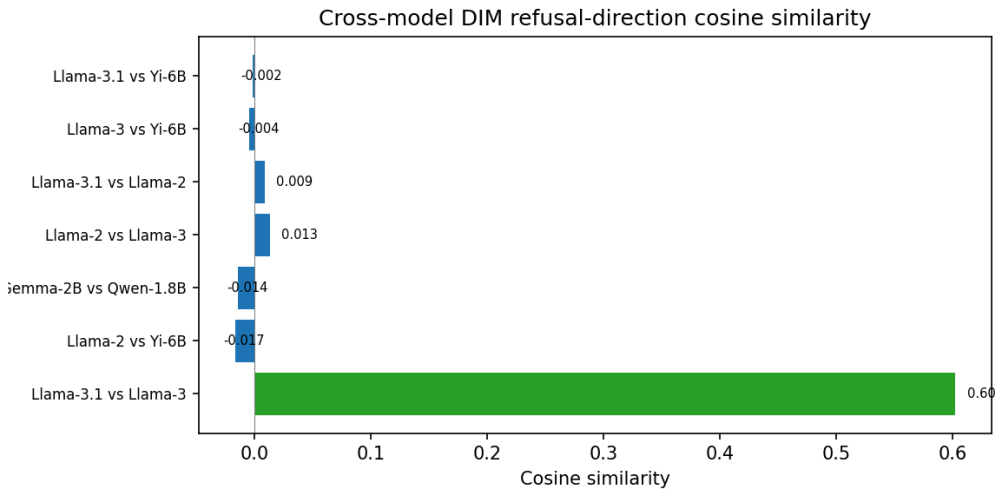

Maximum Subspace Overlap (MSO) [4] measures overlap of the DIM refusal direction with ActSVD's weight-delta subspace per layer and weight type.

Most cells are **near random baseline** ($\approx k_A \cdot k_B/d$) — the two methods find **nearly orthogonal** safety structures.

Only notable signal: **layer 10** MLP down proj (MSO = 0.057, 3.2× random) — adjacent to DIM's source layer (11).

# Findings — Per-Layer MSO & Cross-Model Cosine:




**Left:** Per-layer average MSO (red) vs random baseline (blue). Layer 10 is the only layer clearly above baseline. Layers 20–31 show no signal. **Right:** DIM directions computed independently for 6 models. Llama-3.1 ↔ Llama-3 cosine similarity = **0.603** — all cross-family pairs ≈ 0. The refusal direction is **model-family-specific**.

## Future Extension: Extensions of Current Work

- Further comparisons of the three methods
- More datasets
  - TwinPrompt dataset from TwinBreak [8]

## Future Extension: Additional Techniques

- Differentiated Directional Intervention [9]
  - More advanced version of difference-in-means
- Prompt optimization [10]
  - Avoiding words that activate the harmfulness subspace
- Evaluate Mode Subspace Overlap (MSO) between safety subspaces
- Implement and evaluate further jailbreaking techniques

**Contribution:**

- Evan: Project Management, Paper reimplementation (20%)
- Adam: Report writeup, Model analysis (20%)
- Calvin: Model analysis, Literature review (20%)
- Kyle: Paper reimplementation, Report writeup (20%)
- Zeke: Model analysis, Slide writeup (20%)

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